IMPERIAL COLLEGE LONDON

**EE4-55**
**EE9-AO11**

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

# MEMS AND NANOTECHNOLOGY

Wednesday, 29 April 10:00 am

Time allowed: 3:00 hours

**There are FIVE questions on this paper.**

**Answer Question 1.**

**Answer Question 2 OR Question 3.**

**Answer Question 4 OR Question 5.**

*Question 1 carries 40% of the marks. Remaining questions carry 30% each.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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Second Marker(s) : A.S. Holmes, Z. Durrani, A.S. Holmes

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**Information for Candidates**

| | |
|---|---|
| Permittivity of free space: | $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m. |
| Boltzmann constant: | $k_B = 1.38 \times 10^{-23}$ J/K |
| Electron charge: | $e = 1.6 \times 10^{-19}$ C. |

The fractional change in resistance for a <110>-aligned piezoresistor on a (100) silicon wafer subject to longitudinal and transverse axial stresses $\sigma_L$ and $\sigma_T$ may be written as:

$$\frac{\Delta R}{R} = \sigma_L \pi_L + \sigma_T \pi_T$$

where $\pi_L = \frac{1}{2}\{\pi_{11} + \pi_{12} + \pi_{44}\}$ and $\pi_T = \frac{1}{2}\{\pi_{11} + \pi_{12} - \pi_{44}\}$, and $\pi_{ij}$ are the standard piezoresistive coefficients.

The maximum stress $\sigma_{max}$ in a square membrane of side $L$ and thickness $h$ subject to a uniformly distributed load $p$ is:

$$\sigma_{\max} \approx 0.3\frac{pL^2}{h^2}$$

For an elastic beam that is built in at both ends, the critical axial load $P_C$ at which buckling will occur is given by:

$$P_C = \frac{4\pi^2 EI}{L^2}$$

The lowest order self-resonance frequency of a cantilever is given by:

$$\omega_0 = \frac{3.52}{L^2}\sqrt{\frac{EI}{\rho A}}$$

The equations of motion for a linear vibratory gyroscope are:

$$m\ddot{v}_x + c_x\dot{v}_x + k_x v_x + 2m\Omega\dot{v}_z = F_x$$
$$m\ddot{v}_z + c_z\dot{v}_z + k_z v_z - 2m\Omega\dot{v}_x = F_z$$

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**This question is compulsory**

1. a) Consider a one-dimensional semiconductor quantum well of width $L = 5$ nm. The energy levels in the well may be approximated by:

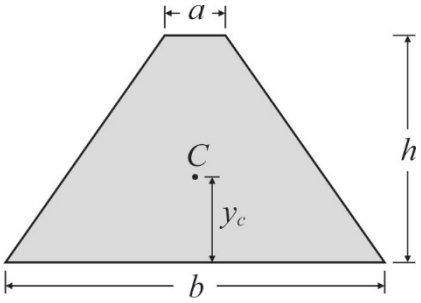
$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

where $\hbar = 1.055 \times 10^{-34}$ J.s is the reduced Planck's constant and $m = 1 \times 10^{-30}$ kg is the electron effective mass. At temperature $T = 300$ K, if there is a 0.5% probability that energy level $E_1$ is occupied by an electron, what is the probability that level $E_2$ is occupied? [7]

b) Assuming the wave function of an electron of mass $m$ travelling in a potential $V$ is given by $\psi = A.\exp[i(kx - \omega t)]$, derive the 1-D time-dependent form of Schrödinger's Equation. [7]

c) A Si MOSFET is scaled in size, such that the channel length $L$, channel width $W$, and gate oxide thickness $t_{ox}$ are all divided by a factor $K > 1$. In order to keep the electric field constant, how should the doping concentrations and the bias voltages be changed? Hence, show that the transverse and longitudinal electric fields in the channel, $F_x$ and $F_y$, remain constant. [6]

d) You are required to calculate the out-of-plane bending stiffness of an anisotropically etched silicon beam fabricated in a (100) wafer. If the dimensions of the beam cross-section are as in Figure 1.1, show that the correct form for the second moment of area in the bending equation is as shown on the right.




$$I = \frac{h^3(a^2 + b^2 + 4ab)}{36(a + b)}$$

*Figure 1.1*

Note that the neutral axis passes through the centroid, $C$, which is at a height $y_c$ above the base of the beam where $y_c = \frac{(b+2a)h}{3(b+a)}$. [6]

**Question 1 continues on the next page.**

**Question 1 continued.**

e) A p-type piezoresistor is to be used to measure the deflection of a square silicon membrane in a bulk micromachined pressure sensor. Sketch a diagram showing where the piezoresistor should be placed for maximum response and derive an approximate expression for the fractional resistance change as a function of the pressure difference across the membrane, the membrane dimensions and the standard piezoresistive coefficients for silicon. [5]

f) A series of straight, built-in beams is to be used to monitor the residual stress of a polysilicon deposition process. The beam lengths are chosen to be 2320, 1640, 1340, 1160, 1038 and 947 μm. If the polysilicon layer thickness is 2 μm, and the three longest beams are found to be buckled after release, what can you say about the residual stress level? [4]

g) A resonant chemical sensor is fabricated from a silicon cantilever with a selectively absorbing polymer layer of thickness 0.5 μm covering its upper surface. The polymer has a density of 1050 kg/m$^3$ and you may assume its Young's modulus is negligible. If the cantilever is 20 μm wide, 200 μm long, and 5 μm deep, estimate the shift in frequency of the lowest order vibrational mode when the polymer absorbs 1 pico-gram of analyte. [5]

Assume values of 170 GPa and 2330 kg/m$^3$ for the Young's modulus and density of silicon.

**End of Question 1.**

2. Figure 2.1 shows a cross-sectional view of an *n*-channel, depletion mode, silicon-on-insulator MOSFET, where the channel is formed by a Si layer of thickness *D* and doping $N_D$, lying on a $SiO_2$ substrate. A top $SiO_2$ layer of thickness $t_{ox}$ forms the gate oxide.

   a) Assuming 'flat-band' conditions at gate voltage $V_g$ = 0 V and $V_{ds}$ = 0 V, sketch the energy bands along the line *AA*' for partial carrier depletion in the *n*-channel. Your diagram should show the conduction and valance band edges $E_c$ and $E_v$ , and the intrinsic level $E_i$, in the Si. The diagram should also show the Fermi energy $E_F$, and the width of the depletion region $W_D$.

   [5]

   b) Sketch the charge density per unit volume $\rho(x)$ along the line *AA'*.

   [5]

   c) Hence, by solving Poisson's equation, $-\frac{\partial^2 V}{\partial x^2} = \frac{\partial F}{\partial x} = \frac{\rho(x)}{\varepsilon_0 \varepsilon_{Si}}$ in the Si channel region, find the potential $V_s$ at the Si – gate oxide interface, at $x = 0$. Here, $\varepsilon_0$ is the permittivity of free space, $\varepsilon_{Si}$ is the relative permittivity of Si, *F* is the electric field and *V* is the potential.

   [15]

   d) Using the result of part c), show that the threshold voltage $V_{th}$, in terms of the Si layer thickness *D* and gate oxide capacitance per unit area $C_{ox}$, is given by:

$$V_{th} = \frac{eN_D D}{C_{ox}} + \frac{eN_D D^2}{2\varepsilon_0 \varepsilon_{Si}}$$

   [5]

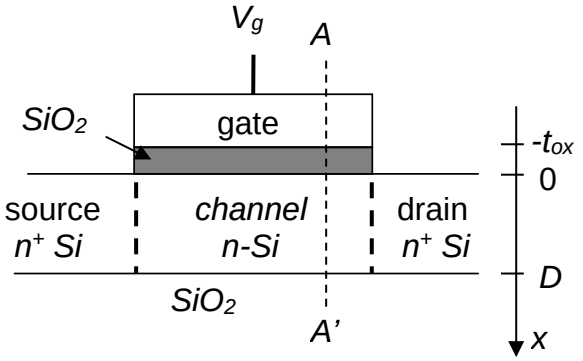



*Figure 2.1*

3. Figure 3.1 shows a one-dimensional potential well, where the potential energy $V(x)$ is given by:

$$V(x) = -V_0 \qquad -a/2 \le x \le a/2$$
$$V(x) = 0 \qquad \text{elsewhere}$$

a) For $-V_0 < E < 0$, write down the general forms of the wave-functions $\psi_1$, $\psi_2$ and $\psi_3$ in the regions $x < -a/2$, $-a/2 \le x \le a/2$, and $x > +a/2$ respectively.

[6]

b) For $-V_0 < E < 0$, write down the wave-vectors $k_1$ and $k_2$ within, and outside, the well respectively.

[4]

c) Hence, show that for $-V_0 < E < 0$, solutions to the Schrödinger equation must satisfy the conditions:

$$\alpha \tan\alpha = \beta \quad \text{or} \quad -\alpha \cot\alpha = \beta$$

where $\alpha$ and $\beta$ are functions of $k_1$ and $k_2$ respectively.

[12]

d) What are the restrictions on these solutions? Hence, show that the wave-functions within the potential well are either odd or even.

[8]

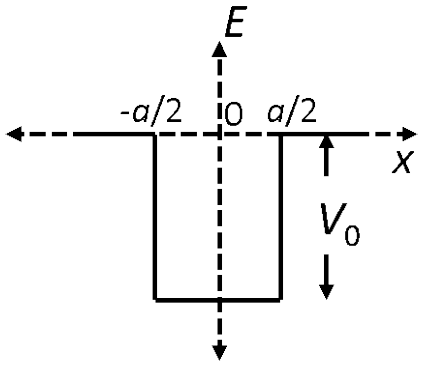



*Figure 3.1*

4. Figure 4.1 shows an open-loop MEMS vibratory gyroscope fabricated in a nickel mechanical layer on a glass substrate. The device is driven in the *x*-axis by a pair of comb-drive electrostatic actuators (not drawn to scale) and sensed in the *z*-axis by means of the parallel plate capacitor formed by the proof mass and an underlying electrode. The mechanical layer is $h$ = 10 µm thick and the gap beneath it is $d$ = 2 µm. The suspension flexures are 450 µm long and 10 µm wide, and the proof mass has an area of 4 mm$^2$. The comb-drive electrodes are 8 µm wide and separated by gaps of $g$ = 2 µm. There are $N$ = 98 gaps in each comb-drive.

   a) Calculate the resonance frequency $\omega_{0x}$ for in-plane vibration assuming values of 8900 kg/m$^3$ and 200 GPa for the density and Young's modulus of nickel. Explain why the resonance frequency $\omega_{0z}$ for out-of-plane motion is expected to be the same. What factors might cause discrepancy between the two resonance frequencies? [8]

   b) The gyroscope is driven differentially by grounding the proof mass and applying voltages $V_A = V_0(1 - \cos\omega_{0x}t)$ and $V_B = V_0(1 + \cos\omega_{0x}t)$ to terminals $A$ and $B$ respectively. Show that the total electrostatic driving force in this case is given by:

$$F_x = \frac{2Nh\epsilon_0}{g} V_0^2 \cos\omega_{0x}t$$

   [6]

   If the Q-factor for in-plane vibration is 1000, calculate the value $V_0$ that will give a vibration amplitude of 10 µm. [4]

   c) Starting from the gyroscope equations of motion, derive an expression for the out-of-plane vibration amplitude when the device is driven as in part b) and subject to a constant rotation rate Ω about the *y*-axis. [4]

   If the Q-factor for out-of-plane vibration is 100, calculate the *z*-axis vibration amplitude when Ω = 0.02 rad/sec. [4]

   d) Estimate the peak-to-peak variation in the sensed capacitance corresponding to the *z*-axis vibration amplitude you calculated in part c). [4]

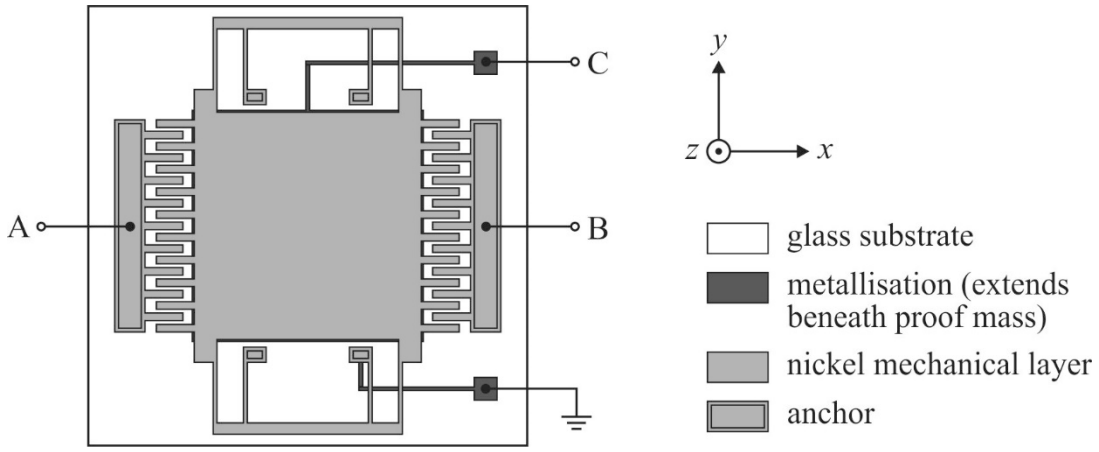



*Figure 4.1*

5. Figure 5.1 shows a normally-open gas valve based on MEMS technology. Gas enters through a hole in the silicon substrate and passes around a circular nickel pad suspended by four flexures. To close the valve, a voltage is applied between the pad and the substrate. A thin oxide layer prevents a short-circuit occurring when the pad is pulled down. Referring to Figure 5.1, the key dimensions are $D_o$ = 3.0 mm, $D_i$ = 1.0 mm, $L$ = 1.5 mm, $w$ = 0.5 mm, $h$ = 10 µm, $g_0$ = 2 µm.

   a) Calculate the stiffness $k$ of the suspension assuming $E$ = 200 GPa for nickel. [6]

   b) The valve is deployed in a setup which ensures the pressure drop $\Delta p$ between inlet and outlet remains constant. Assuming all of this pressure drop occurs in the gap between the pad and the substrate, show that the upward force $F_p$ exerted on the pad by the flowing gas is:

$$F_p = \frac{\pi(D_o^2 - D_i^2)}{8\ln(D_o/D_i)}\Delta p$$

   [8]

   You may assume that the pressure $p$ beneath the pad varies with radius $r$ as:

$$p(r) - p_0 = \begin{cases} \Delta p & 2r < D_i \\ \dfrac{\ln(D_o/2r)}{\ln(D_o/D_i)}\Delta p & D_i \le 2r \le D_o \end{cases}$$

   c) Write down an equation for the total force acting on the pad when a voltage $V$ is applied. You may ignore the effect of the dielectric on the electrostatic force. [6]

   By applying the conditions for snap-down, show that the voltage $V_C$ required to close the valve is:

$$V_C = \sqrt{\frac{32k}{27\pi D_o^2 \varepsilon_0}\left(g_0 + \frac{F_p}{k}\right)^3}$$

   [6]

   d) Using the results in parts a) to c), calculate the value of $V_C$ when $\Delta p$ = 1000 Pa. [4]

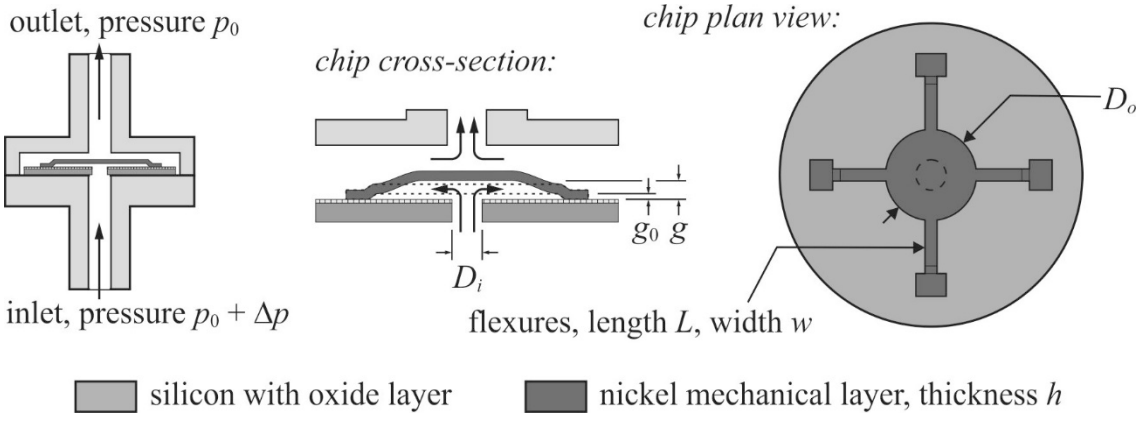



*Figure 5.1*